

Abstract No.

Glioblastoma rewires hematopoiesis and the emerging trajectory strikes back

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We recently discovered hematopoietic stem and progenitor cells (HSPCs) in the tumor tissue of patients with the incurable brain tumor glioblastoma. Here we reveal that a lymphopoietic subpopulation of HSPCs negatively correlates with patient survival and we reveal the underlying cellular mechanism which builds on contact-dependent development of hematopoietic fate in the solid tissue microenvironment.

Through prospective and retrospective analysis of two independent clinical cohorts using patient derived tissue and cellular models for spatial and single cell derived analyses, we were able to decipher the instructive role of hematopoietic evolution on tumor cell activity by specific antigen-receptor interactions. In co-culture models and in co-clinical patient-derived tissue samples, we were able to interfere with this interdependency by pharmacological inhibition of the lymphopoietic trajectory, or by the use of tools that induce antibody-dependent cytotoxicity (ADCC).

Our results highlight a surprising, yet unrecognized role of intra-tumor hematopoietic trajectories that foster glioblastoma growth. An immediate translational route may be considered that makes use of the availability of clinical grade antibodies in settings of controlled clinical trials.



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